

JMO – 2002

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all questions.

1.

(a) Which three numbers have a sum of 200?

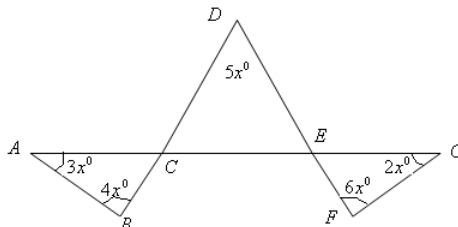
21	28	76	98
85	47	31	15
13	65	39	61

(b) Write $+$, $-$, $\times$ (addition, subtraction, multiplication) signs between the following numbers, so that the equality would be true. $7 \quad 3 \quad 2 \quad 5 \quad 8 \quad = \quad 10.$ (c) a , b , and c are three numbers such that their product is 6581. Find out $a + b + c = ?$

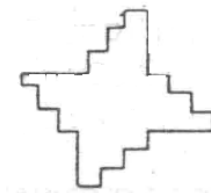
2.

- (a) A's efficiency of doing work is 20% more as compare to that of B. If A can complete a work in 25 days, how many days would B take to complete the same work?
- (b) Three persons A, B and C starts a 10 km race simultaneously in which A beats B by 2 km, and C by 4 km. By how many km does B beat C if their speed is uniform?

3.

(a) .In the given diagram what is the value of x ?

(b) In the diagram adjacent edges are at right angles. The four longer edges are of equal length and all of the shorter edges are also equal in length. The area of the shape is 528. What is the perimeter?

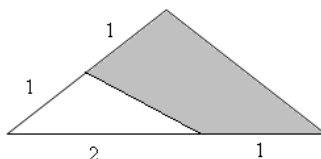


4.

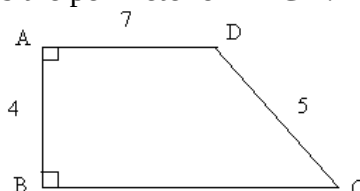
- (a) If the diameter of a circle is 5 units, find the area of an equilateral triangle inscribed in the circle.
- (b) Two barrels contains a certain amount of water. If you pour from the first barrel as much as water into the second barrel as the second barrel had, then you pour from the second barrel as much water into the first barrel as it just had, and finally pour from the first barrel as much water into the second barrel as the second barrel just had, the both barrels contains 160 litre of water. How much water did the barrel contains originally?

- 5.
- (a) Delete some of the digits of the number 82077875072562386 so that remaining number will be the greatest number divisible by 36.
- (b) A defective balance show its beam balanced only when the weight of the material on one pan is of 25% more than the material on the other pan. Without knowing the defect of the balance, a businessman purchased 10 kg of material using the said balance. When he sold the same material using the same balance, he found the material weighing more than 10 kg. The material would have been of what weight while being sold?

6. .
- (a) What fraction of area of largest triangle to that of shaded one?



- (b) Ajay and Bijay must arrive at a town 22.5 km away. They have one bicycle between them and must arrive at the same time. Bijay sets out riding at 8 km/h, leaves the bicycle and then walks at 5 km/h. Ajay walks at 4km/h reaches the bicycle and ride at 10 km/h. For how many minutes was the bicycle not in motion?
- 7.
- (a) The prisoner was released after 6 years and he says: I was 5 times older than my son when I started my prison term, but now I am only 3 times as old he is. When my son gets twice as old as he is now, I will be twice old as him. How old was the prisoner when his son born?
- (b) In the diagram $AD < BC$. What is the perimeter of ABCD?



- 8.
- (a) On an exam with q questions, Marie correctly answered 15 of the first 20 but just $\frac{1}{3}$ of the rest. If the total score was 50%, what was the value of q?
- (b) Solve for x, $2(2^{2x}) = 4^x + 64$.
- 9.
- (a) Two potters went to market to sell their earthen pots. Each of them got as many rupee for each of their pots as many pots they each took to the market. But because their pots were beautiful, they each got as many extra rupees for their pots as many they each took to the market. This way the first potter received Rs 220 more than the second potter. How many pots did they each take to the market?
- (b) Express 144 as sum of two numbers in such a way that HCF of two numbers will be 9.
10. Two pupils of the 5th form and several pupils of the 6th form participated in a chess tournament. Each pupil played once with every other participant. The two pupils of the 5th form together had 8 points, each of the pupils of the 6th form had one and the same number of points (in the tournament a winner receives 1 point a loser receives 0 and for a draw game each of the participants receive $\frac{1}{2}$). How many pupils of the 6th form participated in the tournament?
